From Edge to Refined Intelligence

Connecting a device is no longer enough to call it 'smart'. The real challenge lies in real-time data refining: operating on milliwatts while ensuring total privacy at the edge

Des données orientées Cloud directement depuis les systèmes embarqués
Dati orientati al Cloud direttamente dai dispositivi integrati

Here some critical challenges and opportunities where embedded systems design directly impacts the effectiveness of real-time data engineering at the edge

DATA COLLECTION (SOURCE)

Capturing signals at the point of origin

DATA INGESTION (PIPELINE)

Moving raw bits into system memory

DATA PROCESSING (REFINING)

Transforming noisy waveforms into structured digital features

DATA QUALITY (VALIDATION)

Ensuring data integrity before it ever leaves the silicon

DATA STORAGE (PERSISTENCE)

Efficient local persistence for fault-tolerant streaming

DATA ANALYTICS (INTELLIGENCE)

Shifting analytical compute to the milliwatt level

DATA TRANSFORMATION (SYNTHESIS)

Compressing complex signals into high-value smart payloads

DATA INTEGRATION (TRANSPORT)

Seamlessly connecting Edge nodes to Cloud Data Lakes

DATA GOVERNANCE (LIFECYCLE)

Managing the versioning and health of distributed intelligence

Cloud Integration Map

EDGE FUNCTION	AWS STACK	GCP STACK
Ingestion	IoT Core / Kinesis	IoT Core / Pub-Sub
Processing	Lambda / Glue	Cloud Functions / Dataflow
Analytics	Redshift/Timestream / S3	BigQuery / Cloud Storage
Governance	IoT Device Management	Fleet Management

Refining the Cloud pipeline at the Edge: Intelligence at the point of origin

Sensors & Real-Time Data Processing

HIGH-SPEED DATA STREAMING

Low-latency data pipelines targeting deterministic sub-millisecond latency for real-time, cloud-native processing at the edge

HANDLING RAW BINARY SIGNALS

On-device Data Validation | 99.9% Outlier Filtering

EDGE COMPUTING ARCHITECTURE

Edge-Native Architecture | 90% Bandwidth Reduction via Local Synthesis

Data Efficiency & Payload Optimization

DOWNSAMPLING SMART

Reduce volume without losing entropy

LOCAL AGGREGATION

From raw signals to high-level events

Hardware-Level Security & Data Privacy

DATA ANONYMIZATION

On-chip hashing & encryption

ZERO-CLOUD FOOTPRINT

Zero-Cloud exposure: Raw data stays on-chip, transmitting only refined insights

TinyML: On-Device Data Intelligence

FEATURE ENGINEERING

On-device feature extraction

ANOMALY DETECTION

Pattern recognition on signals

QUANTIZATION

Compressing model weights without losing accuracy

Towards Intelligent Edge

To improve cloud data processes

Which point is critical for your project?

Quel point est critique pour votre projet ?

Quale di questi punti è critico per il tuo progetto ?

@AF